

Task Questions

1. Question 1

Since rebooting the EC2 instance needs administrator level permission, there was an error that says that I'm not the authorized person to perform this operation

2. Question 2

The AWSCloud9User policy that's attached to the AppDevelopers group Nikhil is in.

3. Question 3

The message was, "Connection failed: Access denied for user root"

4. Question 4

The message displayed was, "User Nikhil is not authorized to perform ssm:DescribeParameters"

5. Question 5

Because in the beginning she doesn't have any AWS EC2 Permissions

6. Question 6

The two specific actions in the policy that allow café website on this instance to access the database credentials in the parameter is:

- ssm:GetParameter
- ssm:GetParameters

Quiz

1. Knowledge Check for Module 8

This knowledge check was comprised of the most important questions throughout the module 8. The questions asked was kind of a summary to the whole module 8. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

This knowledge check was mainly about Identity and Access Management (IAM). The first question was about IAM users. Then there were questions about IAM policy, IAM groups, IAM roles. Then there were real life scenario-based questions. Finally, there were questions about identity federation management, about the services it uses.

The screenshot shows the AWS Academy interface for a knowledge check. On the left is a dark sidebar with navigation icons for Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area has a breadcrumb trail: ACav2EN-LTI13-75032 > Assignments > Module 8 Knowledge Check. Below this, the title 'Module 8 Knowledge Check' is displayed. A summary bar shows 'Due' as 'No Due Date', 'Points' as '100', and 'Submitting' as 'an external tool'. The central focus is a box titled 'Module 8 knowledge check results' which contains the following information:

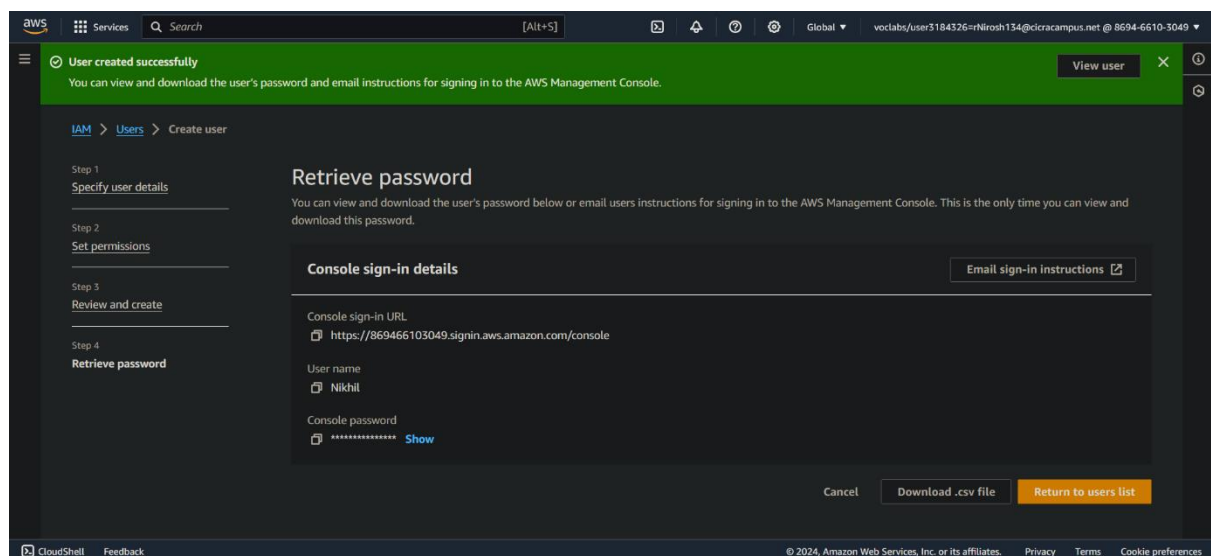
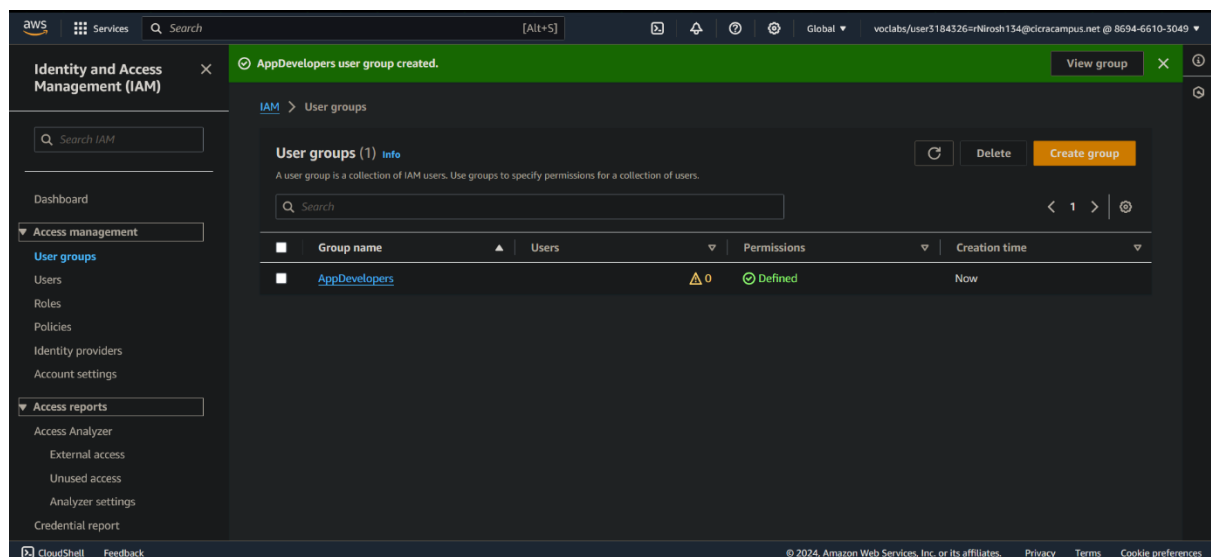
Your score:	80%	(80 points)
Required score:	70%	(70 points)

Result: Congratulations! You have completed this module.
To continue, choose **Next** in the lower-right corner.

Module 8 – Challenge Lab: Controlling AWS Account Access by Using IAM

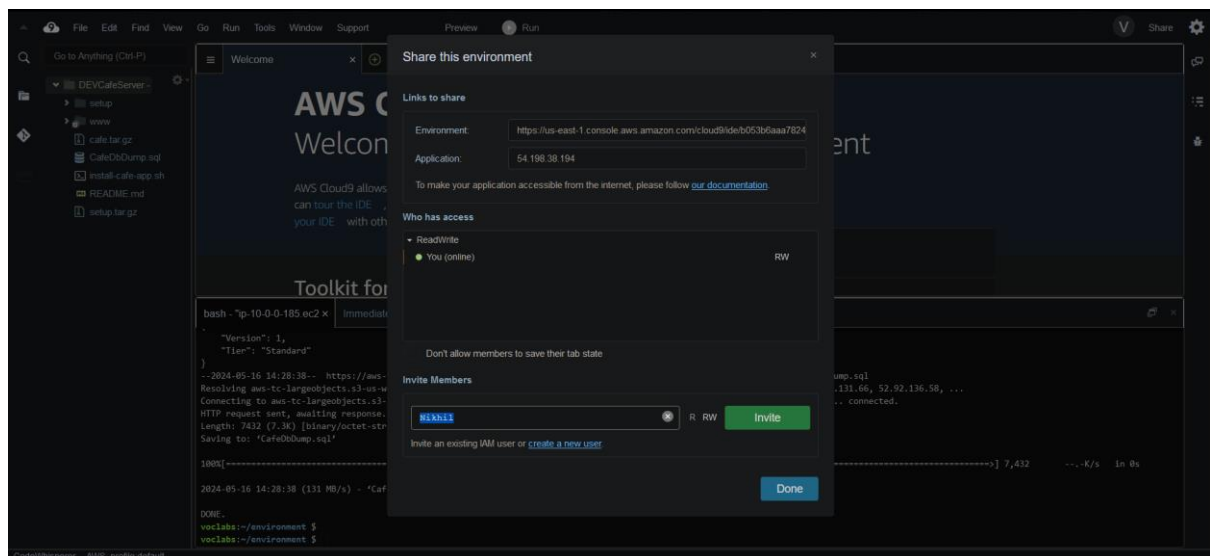
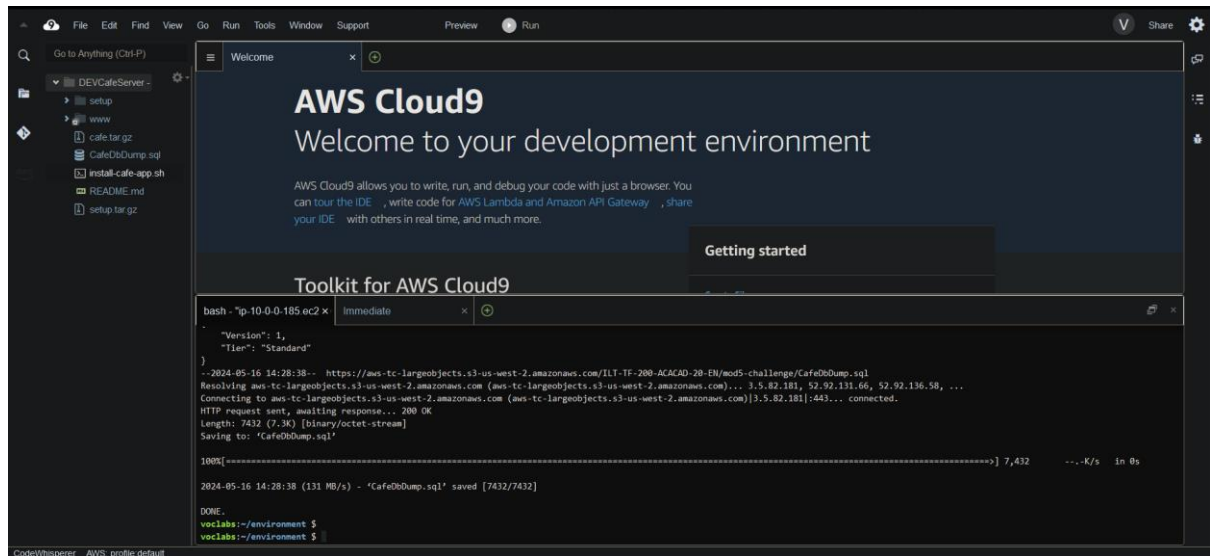
Task 1: Configuring an IAM group with policies and an IAM user

As Sofia, the AWS account administrator, I need to create an IAM group named "AppDevelopers" with specific policies, then create a user named "Nikhil" and add him to this group. After that, I will set up a Cloud9 environment on the DEV Cafe Server EC2 instance by running a setup script. Finally, I will share access to this Cloud9 environment with Nikhil, ensuring he can collaborate on the development tasks.



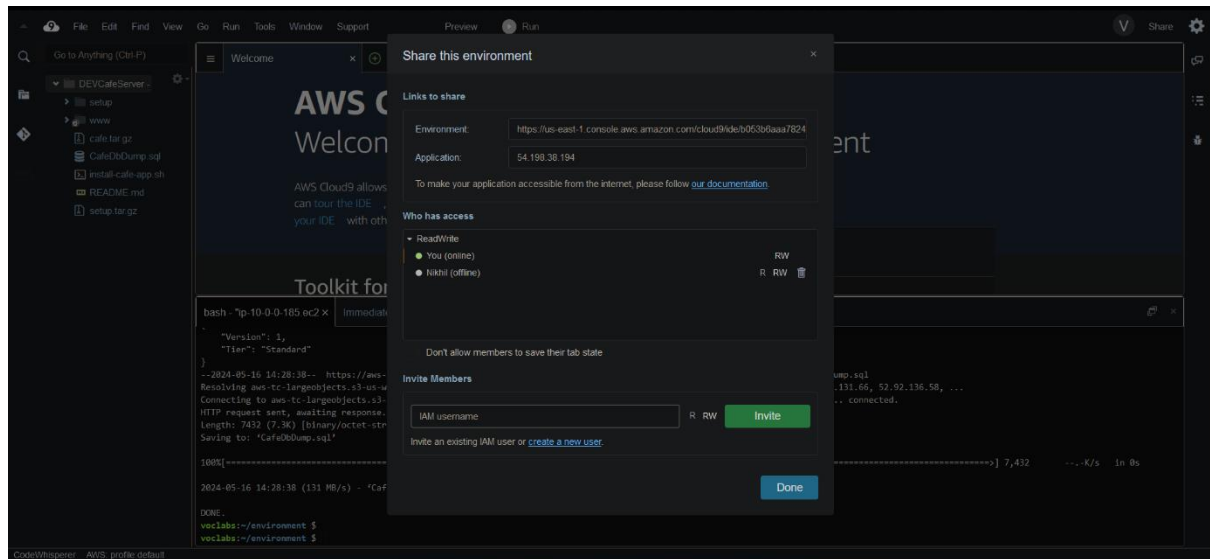
SIT233 – Cloud Computing
Module 8 – Challenge Lab: Controlling AWS Account
Access by Using IAM

Name: Nirosh Ravindran
CICRA Student ID: BSCP|CS|63|134



SIT233 – Cloud Computing
Module 8 – Challenge Lab: Controlling AWS Account
Access by Using IAM

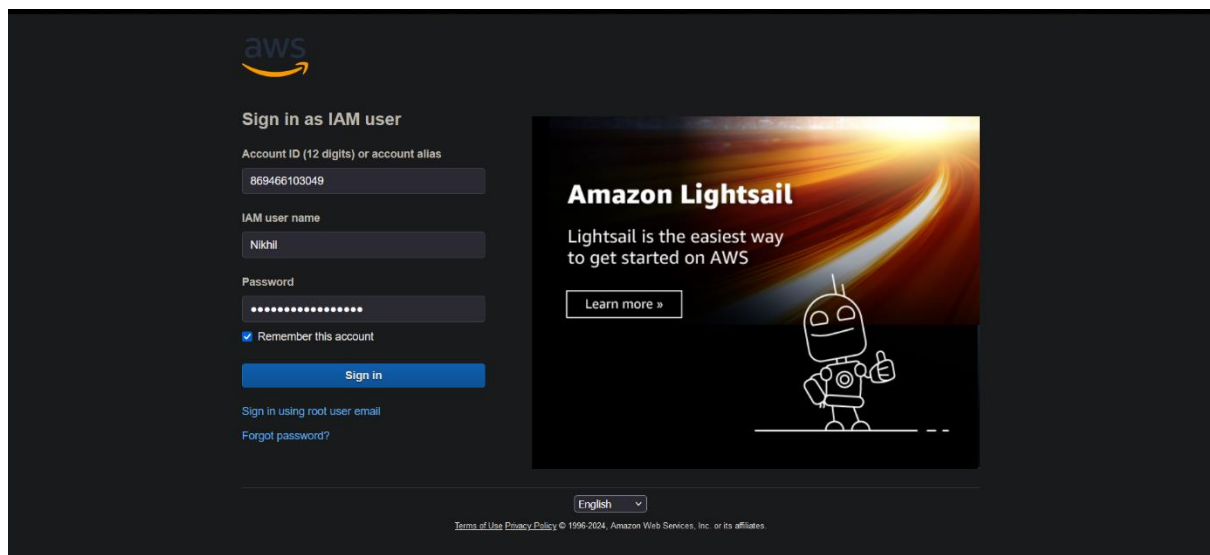
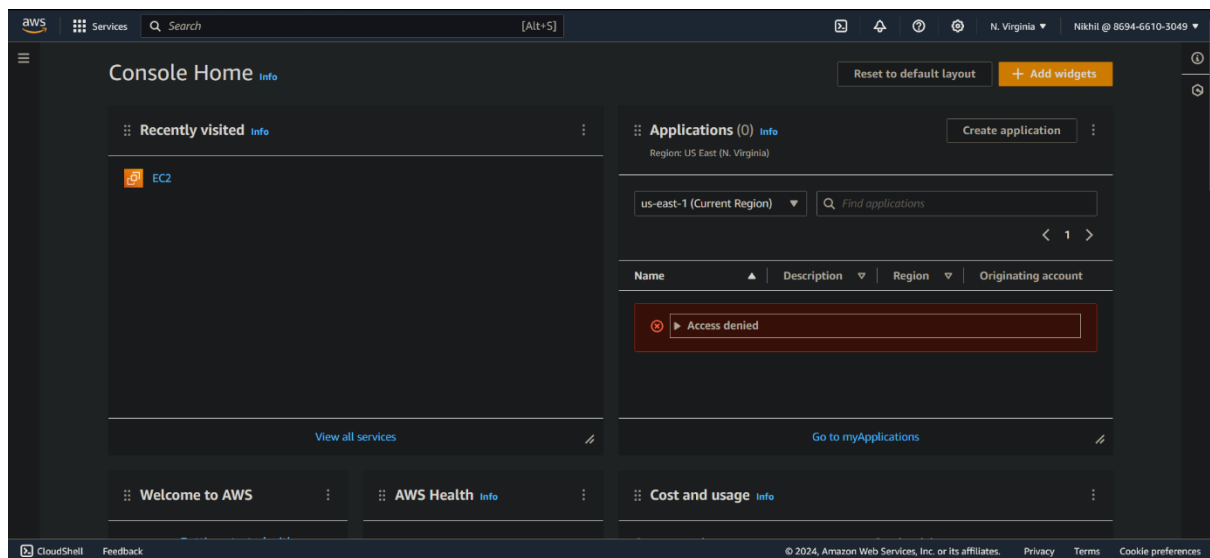
Name: Nirosh Ravindran
CICRA Student ID: BSCP|CS|63|134

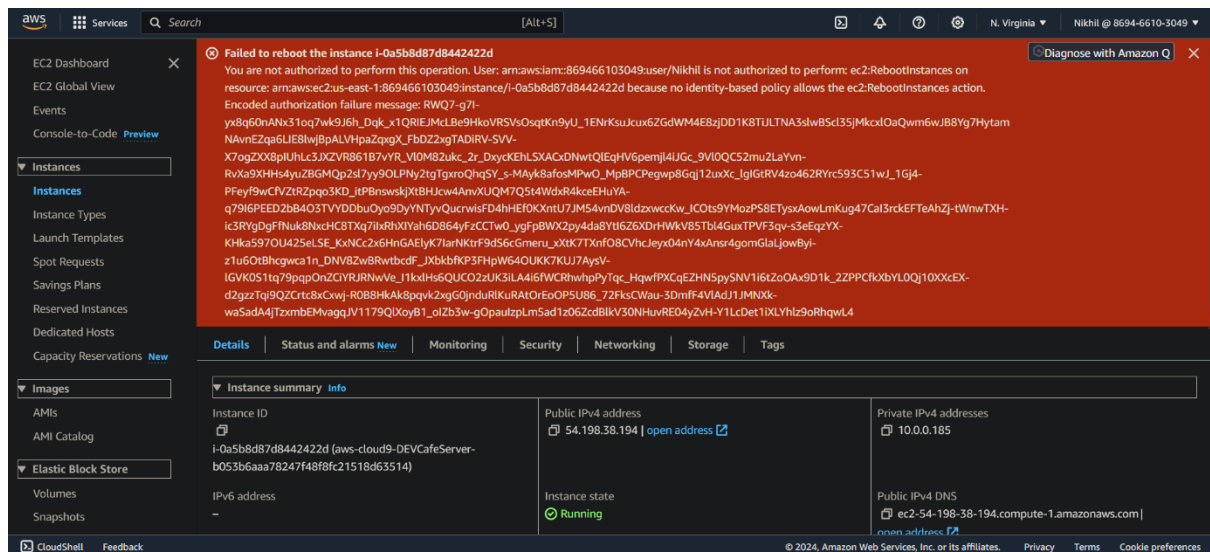
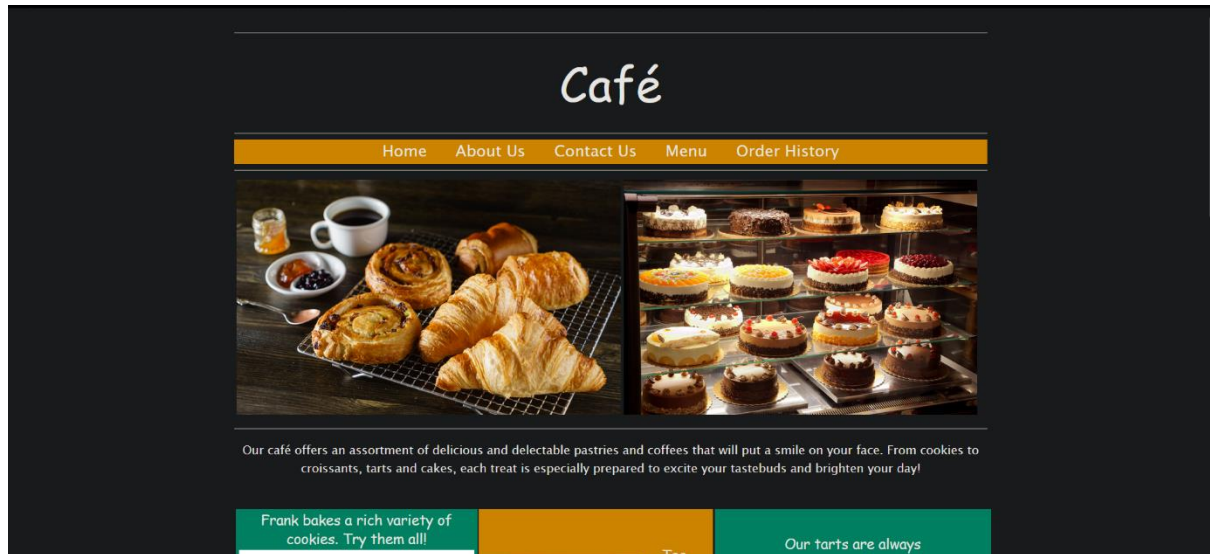


Task 2: Logging in as Nikhil and testing access

As Nikhil, I first log in using an incognito browser tab to keep my session separate from Sofia's. After logging in with Nikhil's credentials, I test my EC2 access by verifying I can view the café web application and attempt to reboot the DEVCafeServer instance.

I answer the lab's multiple-choice questions based on these actions. Next, I access the AWS Cloud9 IDE, edit the café website's index.php file to update the main heading, and check the changes.





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AWS Academy Cloud Architecting - Module 8 Challenge Lab Questions

View questions in: [English](#)

Question 1: What happened when Nikhil tried to reboot the EC2 instance?

- ☐ It rebooted successfully
- ☐ An error indicated that the instance was already stopped, so it should be started instead of rebooted
- ☐ An error indicated that you must first stop the instance
- ☒ An error indicated that you are not authorized to perform this operation

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Question 2: Which IAM policy allowed Nikhil to access the AWS Cloud9 environment?

- ☐ The AWSCloud9EnvironmentMember policy that's attached to the Nikhil IAM user
- ☐ The AWSCloud9EnvironmentMember policy that's attached to the AppDevelopers group Nikhil is in
- ☒ The AWSCloud9User policy that's attached to the AppDevelopers group Nikhil is in
- ☐ The AWSCloud9User policy that's attached to the Nikhil IAM user

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AWS Cloud9 [AWS Cloud9](#) > Environments

My environments
Shared with me
All account environments
Documentation

Environments (1) [Delete](#) [View details](#) [Open in Cloud9](#) [Create environment](#)

Shared with me

Name	Cloud9 IDE	Environment type	Connection	Permissions	Owner ARN
DEVcafeServer	Open	EC2 instance	Secure Shell (SSH)	Read-write	arn:aws:sts:869466103049:assumed-role/voclabs/user3184326-rNirosh134@cicracampus.net

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File Edit Find View Go Run Tools Window Support Preview Run

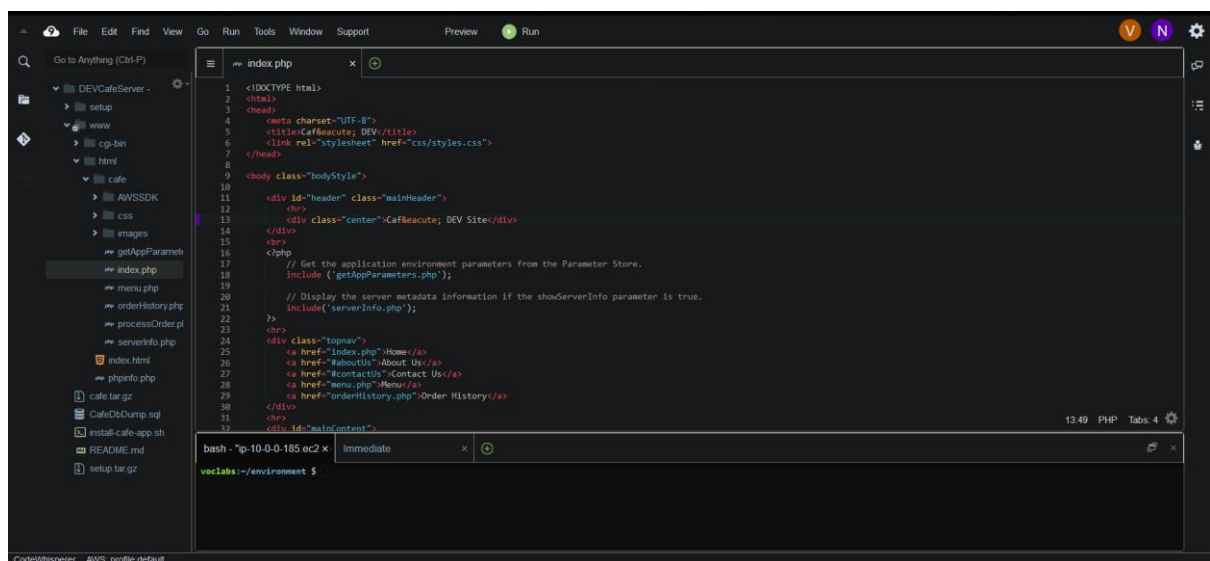
Go to Anything (Ctrl-F)

- DEVcafeServer -
 - setup
 - www
 - cafe.tar.gz
 - CafeDbDump.sql
 - install-cafe-app.sh
 - README.md
 - setup.tar.gz

```
bash - "ip-10-0-0-185-ec2" x Immediate x
voclabs:~/environment $
```

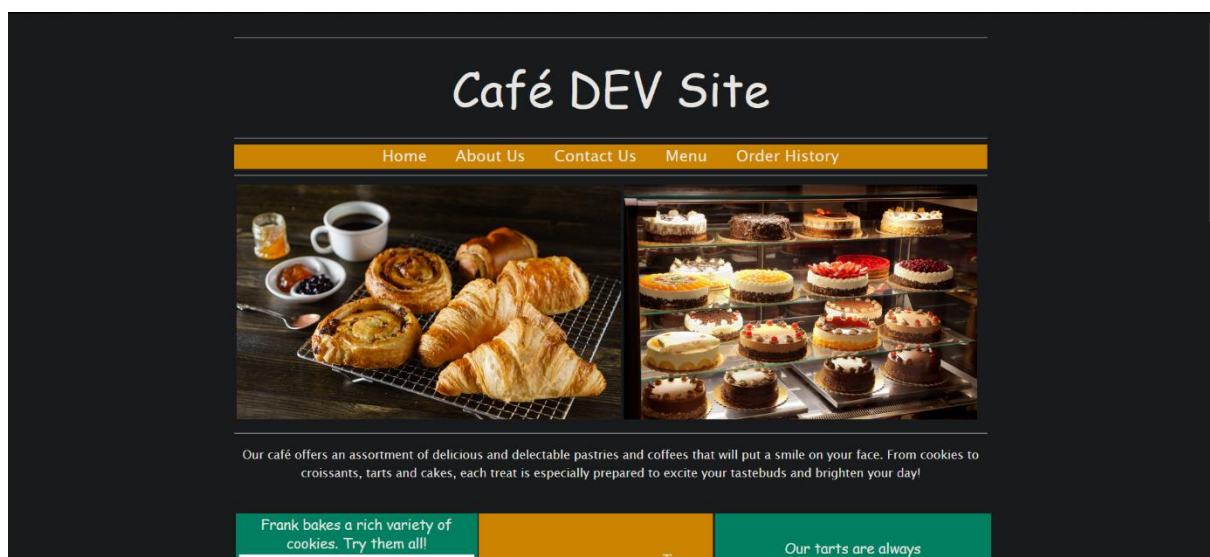

Then I tested the connectivity of the web application's database by viewing the Menu page, noting the displayed message for the lab questions. Additionally, I access the Systems Manager Parameter Store to observe any messages.

Finally, I verify the functionality of the production café website by checking the PRODCafeServer instance, ensuring it operates correctly, and report my findings to Sofía.



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <meta charset="UTF-8">
5 <title>Caféacut; DEV</title>
6 <link rel="stylesheet" href="css/styles.css">
7 </head>
8 <body class="bodyStyle">
9
10 <div id="header" class="mainHeader">
11 <div class="center">Caféacut; DEV Site</div>
12 </div>
13 <div class="center">Caféacut; DEV Site</div>
14 </div>
15 <div class="center">Caféacut; DEV Site</div>
16 </div>
17 // Get the application environment parameters from the Parameter Store.
18 <include ("getAppParameters.php");
19
20 // Display the server metadata information if the showServerInfo parameter is true.
21 <include ("serverInfo.php");
22 </div>
23
24 <div class="topnav">
25 <a href="index.php" Home</a>
26 <a href="aboutUs.php" About Us</a>
27 <a href="contactUs.php" Contact Us</a>
28 <a href="menu.php" Menu</a>
29 <a href="orderHistory.php" Order History</a>
30 </div>
31 <div id="mainContent">
32
```

bash -p 10-0-0-185 ec2 x
voclabs:/environment \$



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Module 8 – Challenge Lab: Controlling AWS Account
Access by Using IAM

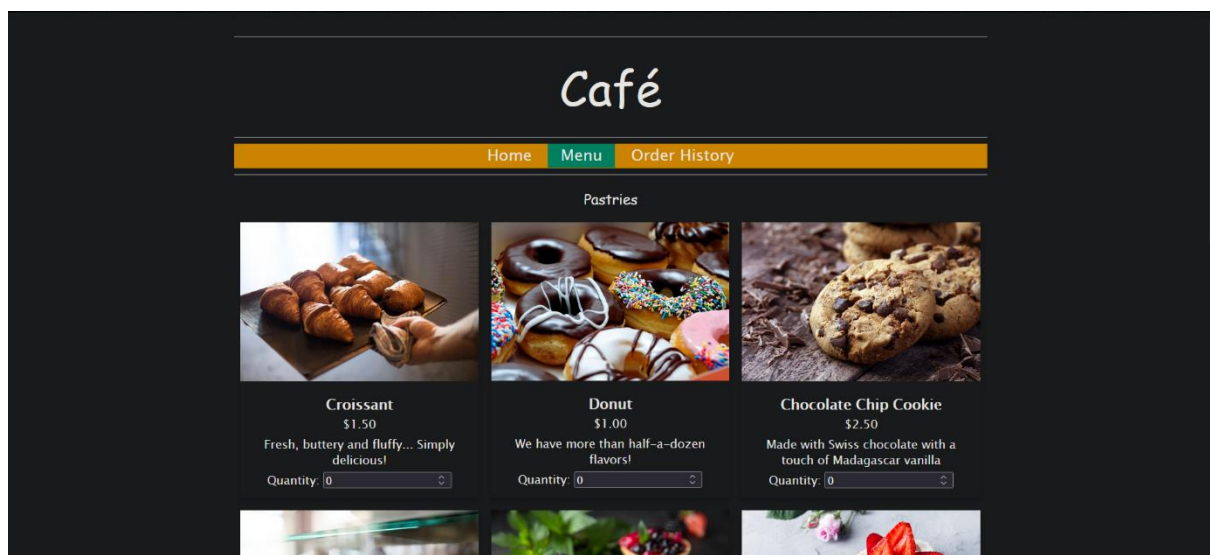
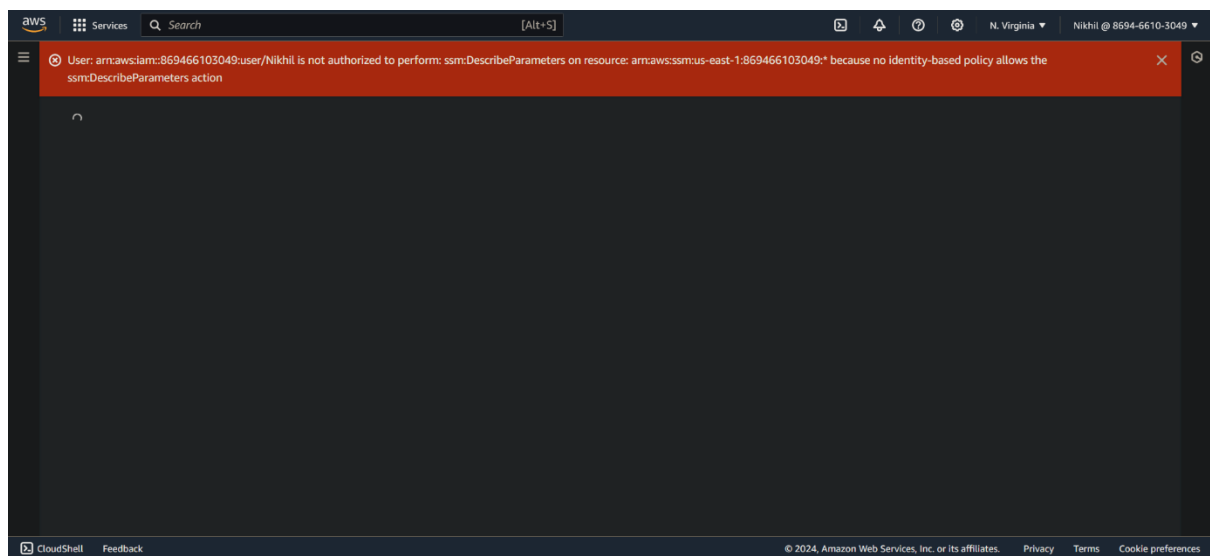
Name: Nirosh Ravindran
CICRA Student ID: BSCP|CS|63|134

Submit

Question 3: Which message was displayed on the Menu page of the cafe website's development instance?

- ☐ Could not contact the database
- ☐ We're sorry, online ordering is not currently available
- ☒ Connection failed: Access denied for user root
- ☐ Error: No results to return

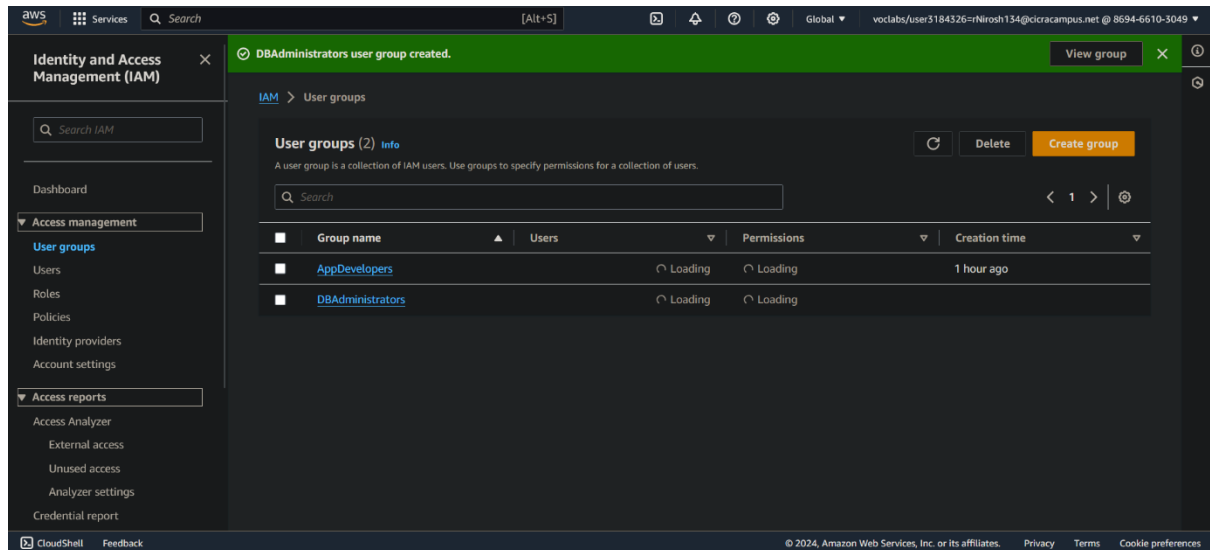
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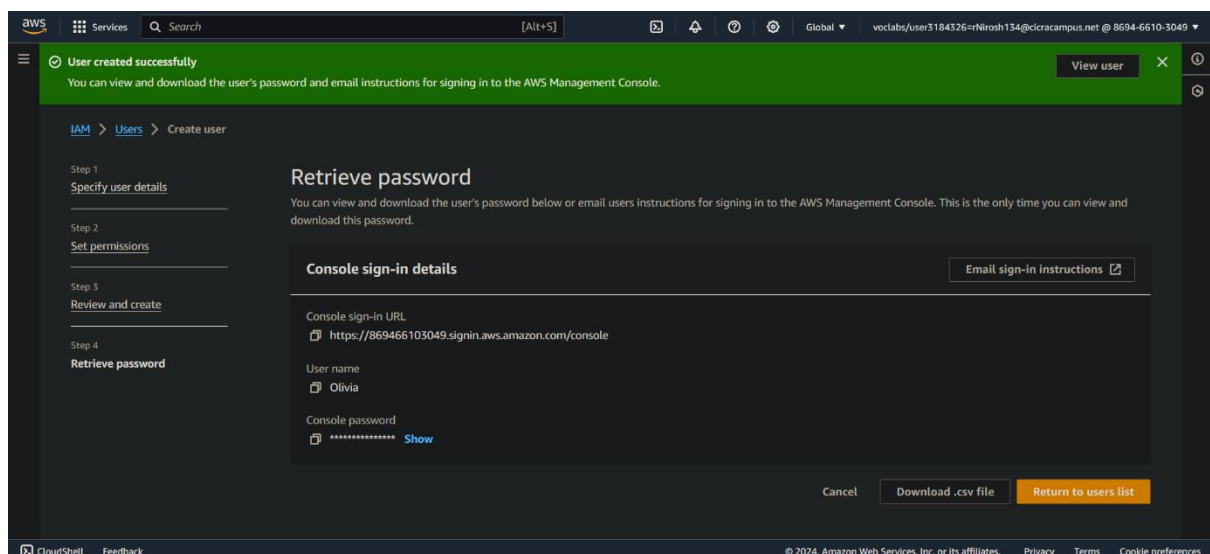


Task 3: Configuring IAM for database administrator user access

As Sofía, I need to set up AWS access for Olivia, the new database administrator. First, I log into the AWS Management Console and create an IAM group called "DBAdministrators," attaching AmazonRDSReadOnlyAccess and AmazonSSMFullAccess policies. Though in a real scenario I would grant more comprehensive RDS access, lab restrictions require using read-only access.



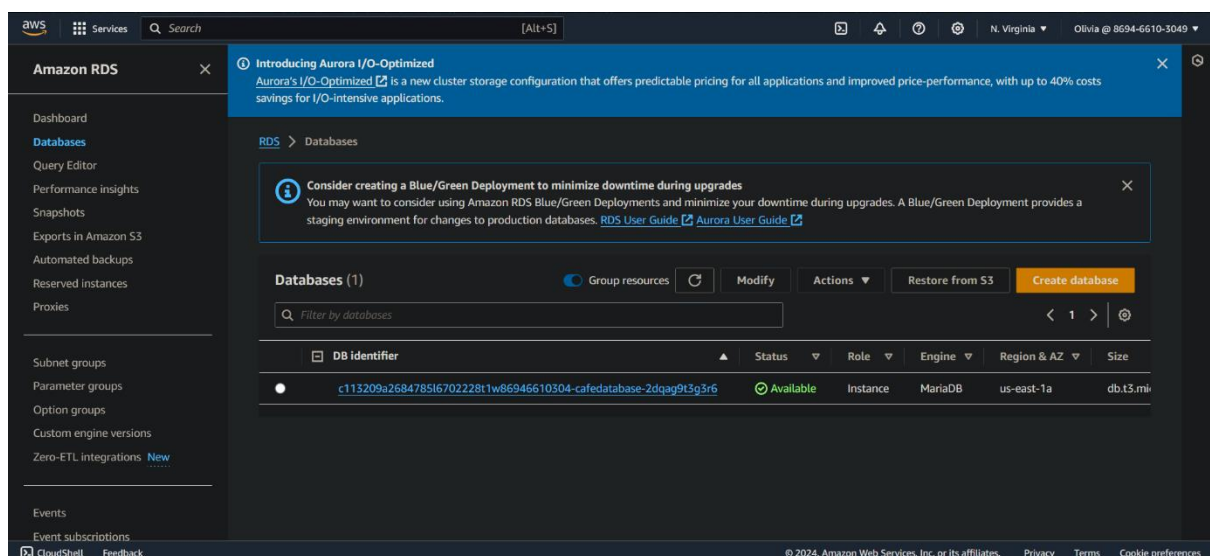
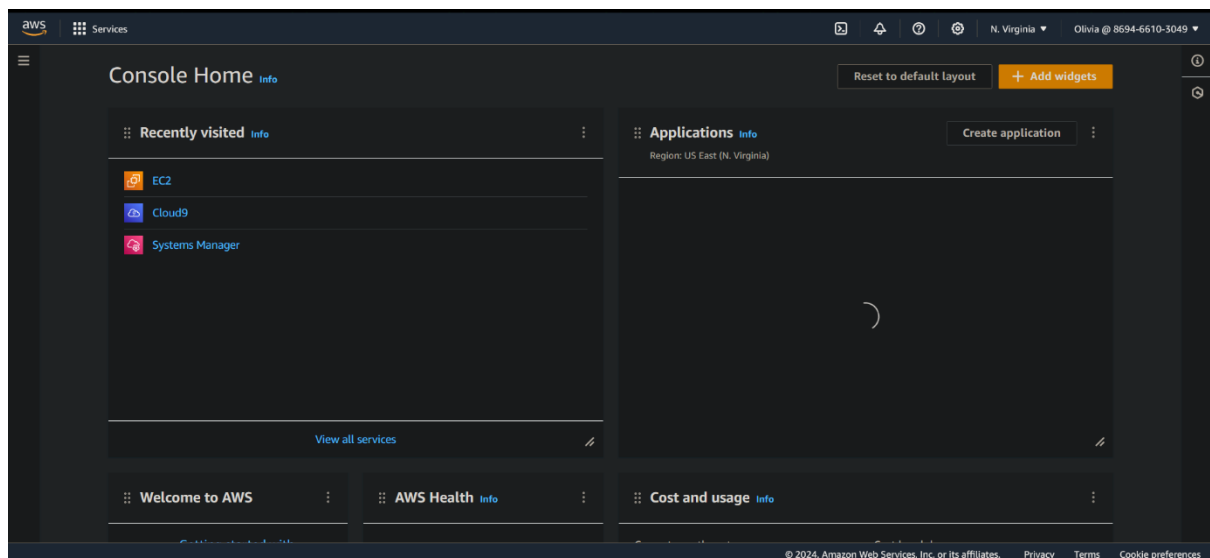
Next, I create an IAM user named Olivia, setting her password to Db@dministrat0r2020! without requiring a password reset. Finally, I add Olivia to the DBAdministrators group, enabling her to access the AWS Management Console with the necessary permissions for her role.



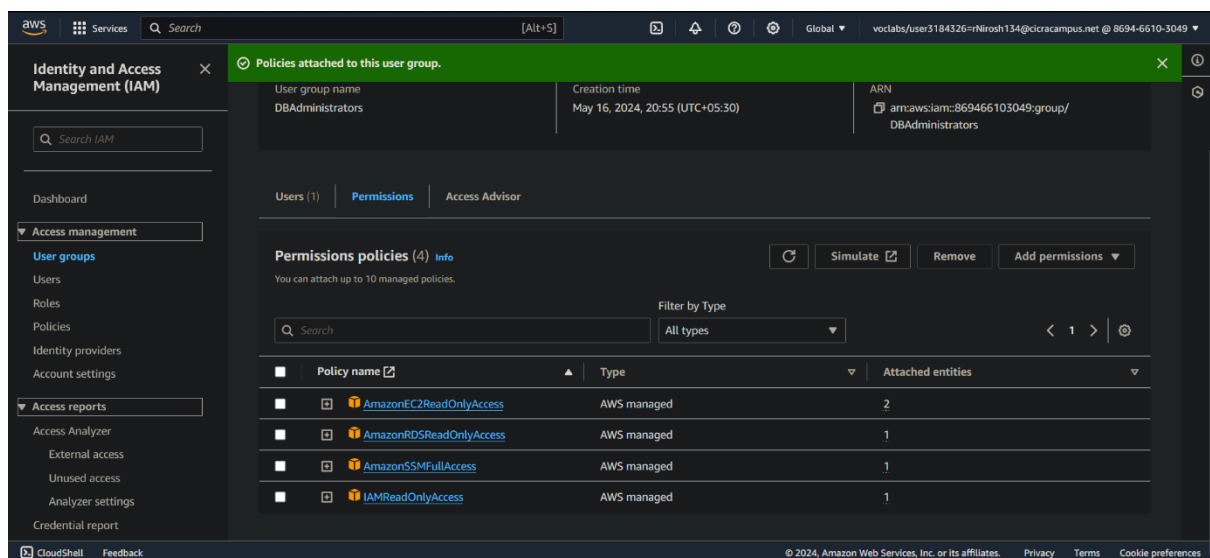
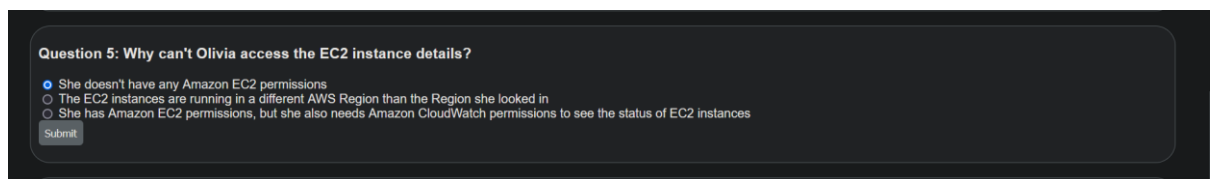
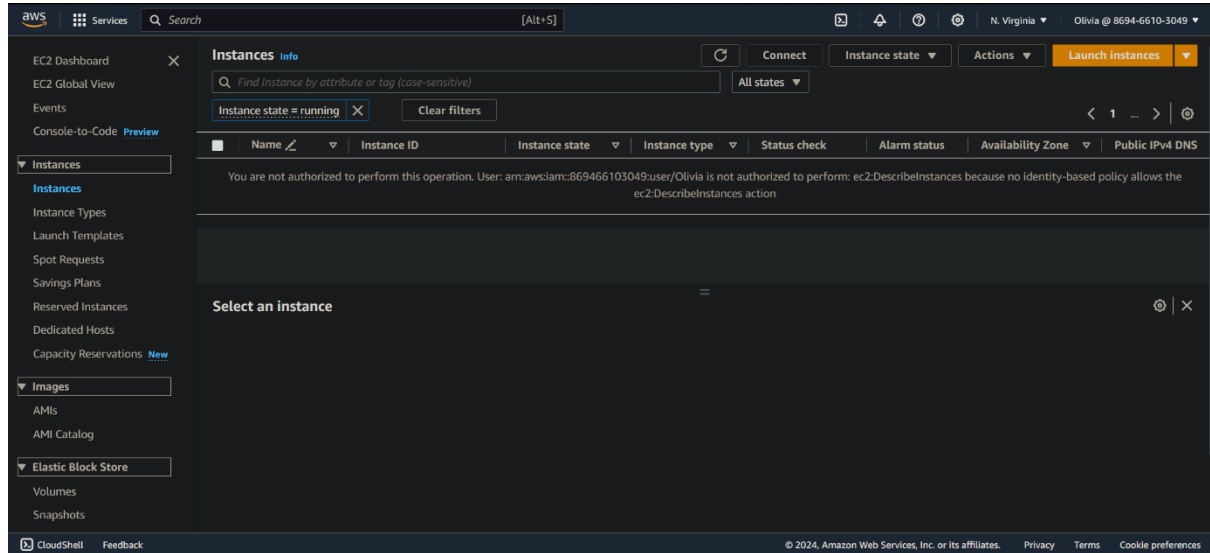
Task 4: Logging in as the database administrator and resolving the database connectivity issue

As Olivia, I log into the AWS Management Console using an incognito browser session. I first verify that the RDS database is running and available. I attempt to access the EC2 instances but realize I lack the necessary permissions, so I inform Sofía.

As Sofía, I update the DBAdministrators group permissions by adding AmazonEC2ReadOnlyAccess and IAMReadOnlyAccess policies.



Returning to Olivia's session, I refresh the EC2 console, access the aws-cloud9-DEV CafeServer instance, and review its IAM role and permissions. I identify that the issue stems from outdated database credentials in the Systems Manager Parameter Store.



I update the dbUser value to the correct RDS credentials, then refresh the café web application to ensure it functions correctly and can submit orders. This resolves the connectivity issue, allowing the development environment to operate smoothly.

The screenshot shows the AWS IAM console 'Allowed services' page. The left sidebar contains navigation links for Identity and Access Management (IAM), Access management, Users, Roles, Policies, Identity providers, Account settings, Access reports, Access Analyzer, External access, Unused access, Analyzer settings, and Credential report. The main content area is titled 'Allowed services (12)' and includes a search bar and a filter dropdown set to 'No Filter'. Below this is a table with three columns: Service, Policies granting permissions, and Last accessed. The table lists 12 services, including Amazon EC2, Amazon EC2 Auto Scaling, Elastic Load Balancing, Amazon RDS, Amazon CloudWatch, Amazon DevOps Guru, AWS Directory Service, Amazon Message Delivery Service, AWS Identity and Access Management, and Amazon CloudWatch Logs. The 'Last accessed' column shows 'Today' for most services and 'Not accessed in the tracking period' for others.

Service	Policies granting permissions	Last accessed
Amazon EC2	AmazonRDSReadOnlyAccess and 2 more	Today
Amazon EC2 Auto Scaling	AmazonEC2ReadOnlyAccess	Today
Elastic Load Balancing	AmazonEC2ReadOnlyAccess	Today
Amazon RDS	AmazonRDSReadOnlyAccess	Today
Amazon CloudWatch	AmazonRDSReadOnlyAccess and 2 more	Not accessed in the tracking period
Amazon DevOps Guru	AmazonRDSReadOnlyAccess	Not accessed in the tracking period
AWS Directory Service	AmazonSSMFullAccess	Not accessed in the tracking period
Amazon Message Delivery Service	AmazonSSMFullAccess	Not accessed in the tracking period
AWS Identity and Access Management	AmazonSSMFullAccess and 1 more	Not accessed in the tracking period
Amazon CloudWatch Logs	AmazonRDSReadOnlyAccess and 1 more	Not accessed in the tracking period

The screenshot shows the AWS IAM console 'Permissions defined in this policy' page. The left sidebar is the same as the previous screenshot. The main content area has tabs for Permissions, Entities attached, Policy versions (2), and Access Advisor. The 'Permissions' tab is active, showing a list of permissions defined in the policy. The permissions are listed in a table with columns for Action, Effect, and Resource. The actions include ssm:DescribeAssociation, ssm:GetDeployablePatchSnapshotForInstance, ssm:GetDocument, ssm:DescribeDocument, ssm:GetManifest, ssm:GetParameter, ssm:GetParameters, ssm:ListAssociations, ssm:ListInstanceAssociations, ssm:PutInventory, ssm:PutComplianceItems, ssm:PutConfigurePackageResult, ssm:UpdateAssociationStatus, ssm:UpdateInstanceAssociationStatus, and ssm:UpdateInstanceInformation. The 'Effect' column shows 'Allow' for all actions. The 'Resource' column shows the ARN: arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore.

Action	Effect	Resource
ssm:DescribeAssociation	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:GetDeployablePatchSnapshotForInstance	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:GetDocument	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:DescribeDocument	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:GetManifest	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:GetParameter	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:GetParameters	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:ListAssociations	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:ListInstanceAssociations	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:PutInventory	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:PutComplianceItems	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:PutConfigurePackageResult	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:UpdateAssociationStatus	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:UpdateInstanceAssociationStatus	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore
ssm:UpdateInstanceInformation	Allow	arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore

Question 6: Name two specific actions in the policy that allow the café web application on this instance to access the database credentials in the Parameter Store.

☐ ssm:ListParameter and ssm:ListParameters

☒ ssm:GetParameter and ssm:GetParameters

☐ ssm:ShowParameter and ssm:ShowParameters

Submit

aws

Services

Search

[Alt+S]

N. Virginia

Olivia @ 8594-6610-3049

Edit parameter request succeeded! View details

AWS Systems Manager

Parameter Store

My parameters

Public parameters

Settings

My parameters

View details

Edit

Delete

Create parameter

Search

Name	Tier	Type	Last modified
/cafe/currency	Standard	String	Thu, 16 May 2024 14:28:34 GMT
/cafe/dbName	Standard	String	Thu, 16 May 2024 14:28:36 GMT
/cafe/dbPassword	Standard	String	Thu, 16 May 2024 14:28:38 GMT
/cafe/dbUrl	Standard	String	Thu, 16 May 2024 14:28:35 GMT
/cafe/dbUser	Standard	String	Thu, 16 May 2024 15:51:25 GMT
/cafe/prodDbUser	Standard	String	Thu, 16 May 2024 14:09:37 GMT
/cafe/showServerInfo	Standard	String	Thu, 16 May 2024 14:28:33 GMT

CloudShell

Feedback

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Café

Home

Menu

Order History

Order Confirmation

Thank for your order! It will be available for pickup within 15 minutes. Your order number and details are shown below.

Order Number: 27

Date: 2024-05-16

Time: 11:56:03

Total Amount: \$1.00

Item	Price	Quantity	Amount
Donut	\$1.00	1	\$1.00

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Task 5: Using the IAM Policy Simulator and creating a custom IAM policy with the visual editor

In this task, I work as Sofía to fine-tune Olivia's IAM permissions. Initially, using the IAM Policy Simulator, I observe Olivia's current access and decide to create a more tailored IAM policy.

In the IAM console, I use the visual editor to craft a policy restricting Olivia's permissions to essential read actions for EC2 and IAM services.

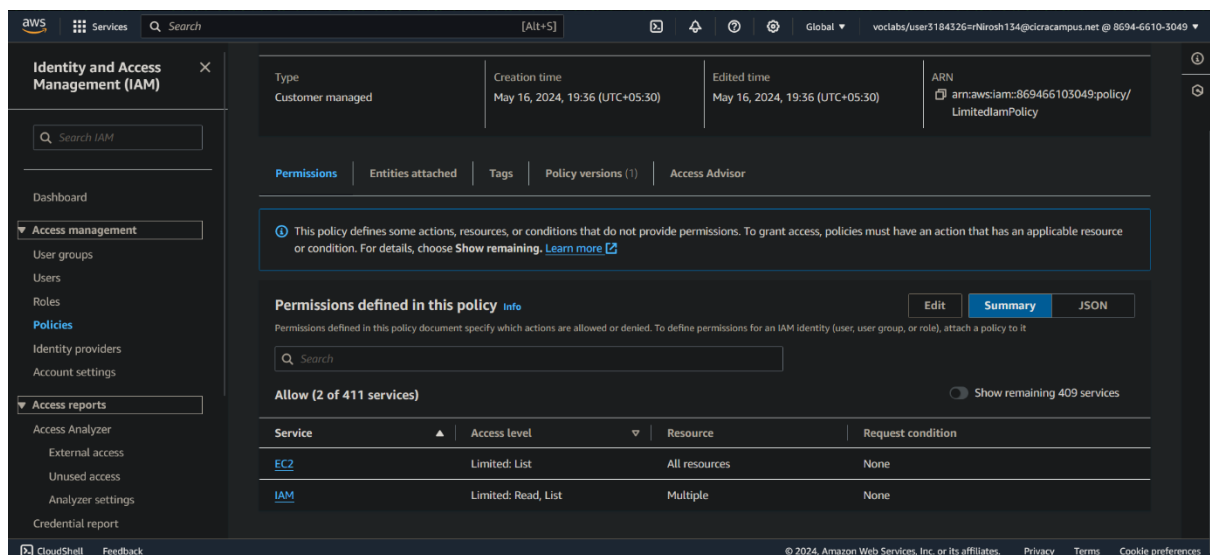
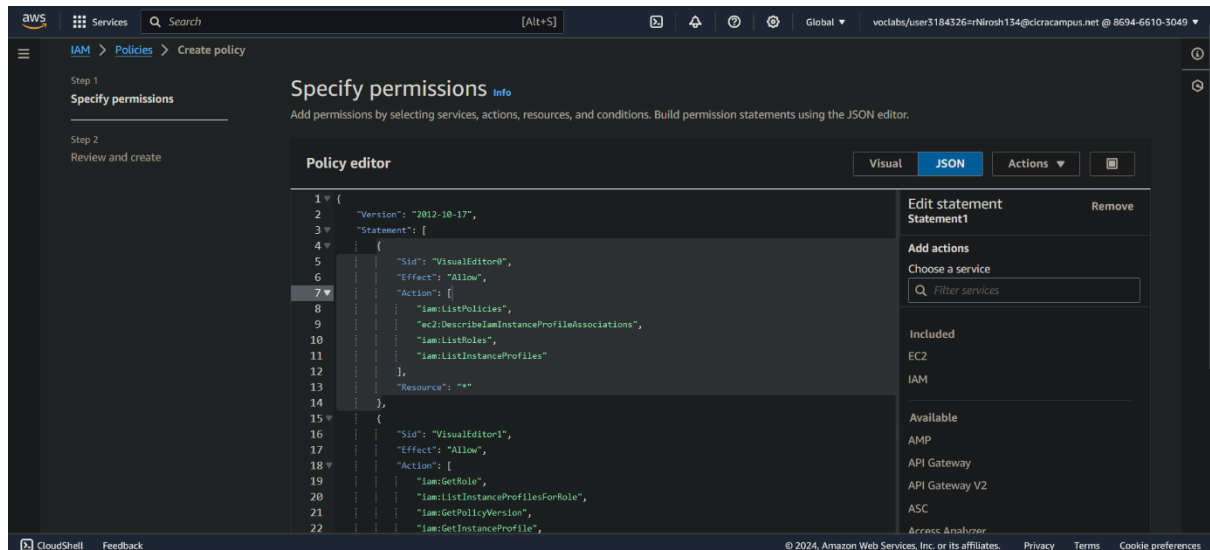
The screenshot shows the IAM Policy Simulator interface. On the left, the 'Policies' sidebar lists 'Selected user: Olivia', 'AWS Organizations SCPs' (checked), and 'IAM Policies' (filtered). The main 'Policy Simulator' pane shows 'Global Settings' and 'Action Settings and Results' with a table header: Service, Action, Resource Type, Simulation Resource, and Permission. The results table is currently empty, indicating 0 actions selected.

The screenshot shows the IAM Policy Simulator interface after selecting 171 actions. The 'Action Settings and Results' table is populated with the following data:

Service	Action	Resource Type	Simulation Resource	Permission
AWS Identity and Access M...	DetachRolePolicy	role	*	denied
AWS Identity and Access M...	DetachUserPolicy	user	*	denied
AWS Identity and Access M...	EnableMFADevice	user	*	denied
AWS Identity and Access M...	GenerateCredentialReport	not required	*	allowed
AWS Identity and Access M...	GenerateOrganizationsAccess...	access-report	*	denied
AWS Identity and Access M...	GenerateServiceLastAccessed...	role,user,group,policy	*	allowed
AWS Identity and Access M...	GetAccessKeyLastUsed	user	*	allowed
AWS Identity and Access M...	GetAccountAuthorizationDetails	not required	*	allowed
AWS Identity and Access M...	GetAccountEmailAddress	not required	*	allowed
AWS Identity and Access M...	GetAccountName	not required	*	allowed
AWS Identity and Access M...	GetAccountPasswordPolicy	not required	*	allowed
AWS Identity and Access M...	GetAccountSummary	not required	*	allowed

However, due to lab constraints, I can't create the policy, but I find a pre-existing one named LimitedIamPolicy that matches my criteria.

I attach this policy to the DBAdministrators group and remove the previous IAMReadOnlyAccess policy.



Upon confirming Olivia's continued access to the necessary IAM role details as Olivia, I ensure her permissions align with her job requirements, granting her the access she needs while reducing unnecessary privileges.

The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Identity and Access Management (IAM), Access management, Access reports, and Access Analyzer. The main content area displays the 'Policies attached to this user group' for the 'DBAdministrators' user group. The policies listed are:

Policy name	Type	Attached entities
AmazonEC2ReadOnlyAccess	AWS managed	2
AmazonRDSReadOnlyAccess	AWS managed	1
AmazonSSMFullAccess	AWS managed	1
LimitedIamPolicy	Customer managed	1

The screenshot shows the AWS IAM console interface with the 'AmazonSSMManagedInstanceCore' policy selected. The policy details are displayed, including the JSON policy document:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "ssm:DescribeAssociation",
8         "ssm:GetDeployablePatchSnapshotForInstance",
9         "ssm:GetDocument",
10        "ssm:DescribeDocument",
11        "ssm:GetManifest",
12        "ssm:GetParameter",
13        "ssm:GetParameters",
14        "ssm:ListAssociations",
15        "ssm:ListInstanceAssociations",
16        "ssm:PutInventory",
17        "ssm:PutComplianceItems",
18        "ssm:PutConfigurePackageResult",
19        "ssm:UpdateAssociationStatus",
20        "ssm:UpdateInstanceAssociationStatus",

```

The policy is attached to the 'AttachRoleToInstance' and 'ReadParameterStore' roles, both of which are customer inline roles with 0 attached entities.

The screenshot shows the IAM Policy Simulator interface. On the left, there are sections for Policies, AWS Organizations SCPs, IAM Policies, and Permissions Boundary Policy. The main area is titled 'Policy Simulator' and shows '171 Action(s) s...'. Below this, there is a table titled 'Action Settings and Results' with columns: Service, Action, Resource Type, Simulation Resource, and Permission. The table lists various actions from AWS Identity and Access Management, all of which are marked as 'denied' with the reason 'Implicitly denied (no matching sta...)'.

Service	Action	Resource Type	Simulation Resource	Permission
AWS Identity and Access M...	AddClientIDToOpenIDConnect...	oldc-provider	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	AddRoleToInstanceProfile	Instance-profile	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	AddUserToGroup	group	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	AttachGroupPolicy	group	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	AttachRolePolicy	role	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	AttachUserPolicy	user	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	ChangePassword	user	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	CreateAccessKey	user	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	CreateAccountAlias	not required	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	CreateGroup	group	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	CreateInstanceProfile	Instance-profile	*	denied Implicitly denied (no matching sta...
AWS Identity and Access M...	CreateLoginProfile	user	*	denied Implicitly denied (no matching sta...

Grade:

My grade is 29/30.

The screenshot shows the AWS Academy challenge lab interface for 'Module 8 Challenge Lab - Controlling AWS Account Access by Using IAM'. The interface includes a sidebar with navigation options like Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main area displays the lab title, due date, points, and submission status. A 'Submit' button is visible. Below this, there is a section titled 'Submitting your work' with instructions and a 'Launch Terminal' button. On the right, a 'Grades' section shows a 'Total score' of 29/30 and a list of answers with their scores.

Submitting your work

50. At the top of these instructions, choose Submit to record your progress and when prompted, choose Yes.

51. If the results don't display after a couple of minutes, return to the top of these instructions and choose Grades

Tip: You can submit your work multiple times. After you change your work, choose Submit again. Your work will be saved to submit again.

Grades

Answer	Score
[Answer 01]	1/1
[Answer 02]	0/1
[Answer 03]	1/1
[Answer 04]	1/1
[Answer 05]	1/1